ECE 5360

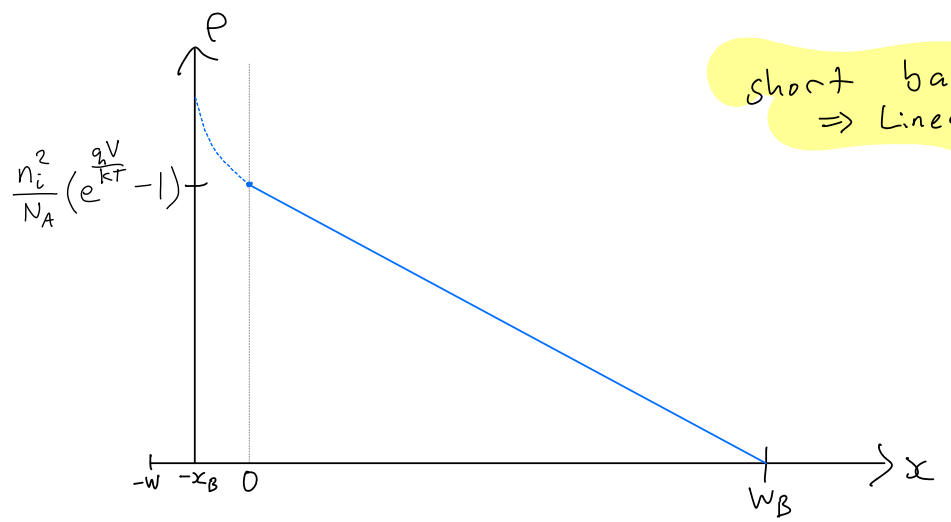
Quiz 3 | Using short base (linear) approximation.

a) Plot of minority carriers in base.

$\Delta n_p(0) = n_{po}\left(e^{qV/kT} - 1\right) \Rightarrow$ By short base approx,

$$\Delta n_p(x) = n_{po}\left(e^{qV/kT} - 1\right)\left(1 - \frac{x}{W_B}\right) \; ; \quad n_{po} = n_i^2/N_A$$

↳ -1 disappears for high V.



short base
$\Rightarrow$ Linear

Now calculate total excess minority charge in quasi-neutral region of base

$\Rightarrow$ Triangle. By geometry,

$$\frac{Q_{hn}}{q} = \int_0^{W_B} \Delta n_p(x)\,dx = \frac{1}{2} \cdot W_B \cdot \frac{n_i^2}{N_A}\left(e^{\frac{qV}{kT}} - 1\right)$$

$$\Rightarrow \boxed{\frac{Q_{hn}}{q} = \frac{W_B n_i^2}{2 N_A}\left(e^{\frac{qV}{kT}} - 1\right)}$$

→ total excess minority carriers (count) in base.

b) Per Eq 5.3.19, for short base,

$$J_t = q n_i^2 \left(\cancel{\frac{D_p}{N_D W_B}} + \frac{D_n}{N_A W_B} \right) \left(e^{qV/kT} - 1 \right)$$

0 ignore emitter term

$$J_t = q n_i^2 \left(\frac{D_n}{N_A W_B} \right) \left(e^{qV/kT} - 1 \right)$$

Textbook Fig 5.5 labels p-side quasi-neutral region "W_E". Problem defines base (p-side) quasi-neutral as W_B. J equation changed accordingly

$$I = JA = \boxed{I = \frac{A q n_i^2 D_n}{N_A W_B} \left(e^{qV/kT} - 1 \right)}$$

It's okay to neglect emitter hole injection since it is a n^+-p contact and $N_D >> N_A$. As such, there is a high electron density once a hole enters the emitter, so any injected holes will quickly recombine, and therefore their contribution to overall current will be minimal.

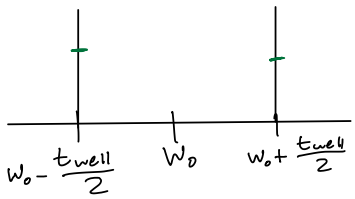
c) Per question, R_{well} is <u>total</u> recombination rate inside well.

has units $cm^{-3} \cdot s^{-1} \Rightarrow J = R_{well} A \cdot t_{well} \cdot q$

Current from recombination

Recombination current: $I_r = J_r \cdot A = \boxed{I_r = q A t_{well} R_{well}} \Rightarrow \frac{C}{s} = A$ ✓

d) $\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + (G - R)$



Like HW MKC 5.10

$$0 = \frac{1}{q}\frac{\partial J_n}{\partial x} - R_{well}$$

Since $t_{well} << W$ & $<< W_B$ (which is already a short base),

let's approximate the well as infinitesimally small

Let $t_{well} \approx 0$, since much shorter than short base.

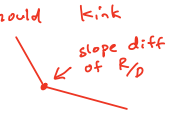
$$\Rightarrow R_{well} = q\frac{\partial J_n}{\partial x} = D_N \frac{\partial^2 \Delta n_p}{\partial x^2} \Rightarrow \Delta n_p = \frac{R}{D_N} = \left.\frac{\partial \Delta n_p}{\partial x}\right|_{W_0 - \frac{t_{well}}{2}} - \left.\frac{\partial \Delta n_p}{\partial x}\right|_{W_0 + \frac{t_{well}}{2}}$$

Let n_0' be before W_0 and n_1' be after $x = W_0$,

$$n_0' = A_0 x + A_1, \quad n_1' = B_0 x + B_1$$

$$n_0'(0) = n_{p0}\left(e^{qV/kT} - 1\right), \quad n_1'(W_B) = 0, \quad n_0'(W_0) = n_1'(W_0), \quad \left.\frac{\partial n_1'}{\partial x}\right|_{W_0} - \left.\frac{\partial n_0'}{\partial x}\right|_{W_0} = \frac{R}{D}$$

↳ Means line should kink



$$\Rightarrow A_1 = n_{p0}\left(e^{qV/kT} - 1\right) \qquad B_1 = -B_0 W_B, \qquad B_0 - A_0 = \frac{R}{D}$$

$$\Rightarrow n_0' = A_0 x + n_{p0}\left(e^{\frac{qV}{kT}} - 1\right) \qquad n_1' = B_0(x - W_B)$$

$$A_0 W_0 + A_1 = B_0(W_0 - W_B) \Rightarrow -W_0\frac{R}{D} + A_1 = -B_0 W_B \Rightarrow B_0 = \frac{W_0 R}{W_B D} - \frac{A_1}{W_B} \Rightarrow A_0 = \left(\frac{W_0}{W_B} - 1\right)\frac{R}{D} - \frac{A_1}{W_B}$$

$$\Rightarrow n_1' = -\left(\frac{D n_{p0}\left(e^{qV/kT} - 1\right) - W_0 R}{W_B D}\right)(x - W_B) = \left(n_{p0}\left(e^{qV/kT} - 1\right) - \frac{W_0 R}{D}\right)\left(1 - \frac{x}{W_B}\right)$$

At W_0, $\Delta n_p(W_0) = \left(n_{p0}\left(e^{qV/kT} - 1\right) - \frac{W_0 R}{D}\right)\left(1 - \frac{W_0}{W_B}\right)$

$$\boxed{\Delta n_p(W_0) = \left(n_{p0}\left(e^{qV/kT} - 1\right) - \frac{W_0 R_{well}}{D_n}\right)\left(1 - \frac{W_0}{W_B}\right)} \qquad \boxed{n_{p0} = \frac{n_i^2}{N_A}}$$

If $R \to 0$, $\Delta n_p(W_0) = n_{p0}\left(e^{qV/kT} - 1\right)\left(1 - \frac{W_0}{W_B}\right)$

This is equivalent to the original diode (part a, in orange)

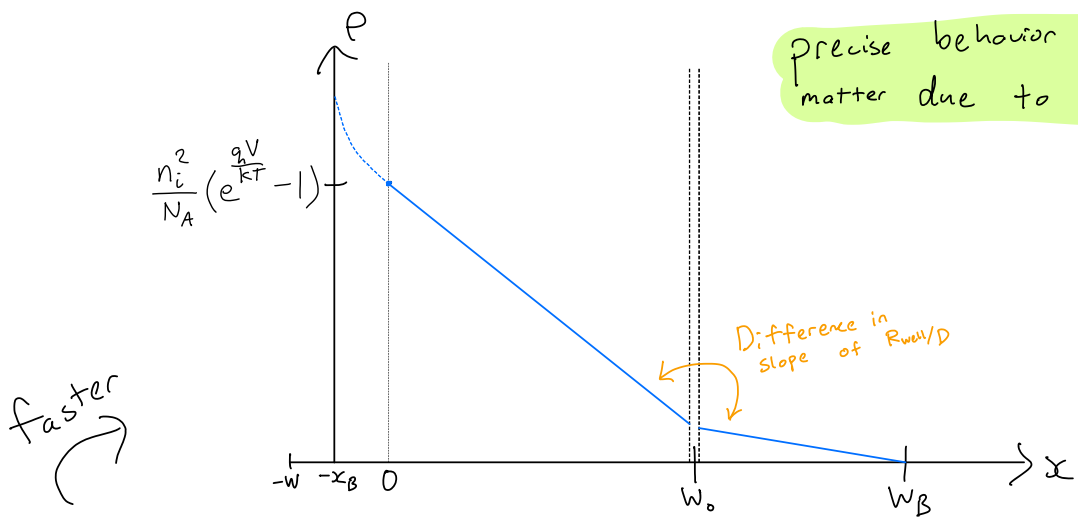
Sanity check ✓

e) From above, $\Delta n_p(W_o) = \left(n_{po}(e^{qV/kT}-1) - \frac{W_o R_{well}}{D_n}\right)\left(1-\frac{W_o}{W_B}\right)$

Clearly, increasing R_{well} decreases $\Delta n_p(W_o)$

Since $\boxed{\Delta n_p(W_o) \propto -R_{well}}$ (highlighted in orange)

Increasing R_{well} makes the coefficient $\left(n_{po}(e^{qV/kT}-1) - \frac{W_o R_{well}}{D}\right)$ smaller



precise behavior inside well doesn't matter due to $t_{well} \to 0$ approximation

The second diode with the quantum well will switch faster.

Clearly, there is less charge stored than in the first diode due to the quantum well kinking the graph. This means #2 has a narrower peak, meaning less stored charge and fewer minority carriers to clear out. This means shorter recovery time.

f) $J = q_h D_n \frac{dn}{dx} \Rightarrow I = -q_h A D_n \frac{dn}{dx} = q_h A D_n \frac{1}{W_B}\left(n_{po}(e^{qV/kT}-1) - \frac{W_o R_{well}}{D_n}\right)$

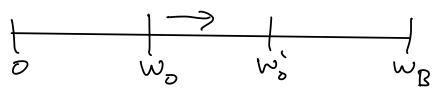
reintroduce. t_{well}

$$\boxed{I = \frac{q_h A D_n n_i^2}{N_A W_B}(e^{qV/kT}-1) - q_h A t_{well} R_{well}}$$

This is **smaller** than the first diode. This is because the carriers recombine earlier due to the well, meaning they do not travel as far into the base (get trapped in well early, decreasing their contribution to current). As such, these these carriers can not contribute as much to the current as in the first diode.

$\Rightarrow$ Smaller current.

g) $\boxed{\text{Move it right}}$



By moving it closer to the end of the base (right), the quantum well will capture carriers later. This allows more carriers to travel farther into the base. Traveling farther before recombining means these carriers will contribute more to the current, increasing overall current.

Right to allow carriers to travel farther before recombining